

Supplement Material

Plasma Protein Risk Scores for Mild Cognitive Impairment and Alzheimer Disease in the Framingham Heart Study

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Running Title: Development of plasma protein risk score analysis for risk of AD

1 sMethod

1.1 Statistical analyses

The least absolute shrinkage and selection operator (LASSO) reduces the regression coefficients to zero by penalizing the regression model using a penalty term known as the L1 norm. The L1 penalty subtracts a multiple of the sum of the absolute coefficients and the tuning parameter from log-likelihood, effectively reducing some regression coefficients to zero. The tuning parameter was optimized using 10-fold cross-validation. The LASSO penalization procedure was applied 100 times to reduce the variability introduced by the 10-fold cross-validation, and the regression coefficients for each protein were obtained as an average of over 100 repetitions. Separate PPRSs for MCI (including CN and MCI individuals) and AD (including CN and AD individuals) were calculated by adding the linear combination of the regression coefficients and their corresponding protein levels.

LASSO selected different numbers of proteins for every run. As a result, we selected the proteins that LASSO selected the most, which is close to 95 times, and have the highest prediction (C-index) with the least proteins. We plotted the differences between the C-index and the number of proteins included in the PPRSs to determine the ideal number of proteins in the final risk scores. **Figure S2** reveals that 36 proteins in MCI have an ideal C-index of 82.5%, while 50 AD proteins have a C-index of 88.15%. The differences in C-indexes tend to be zero as the number of proteins increases after 36 MCI proteins and 50 AD proteins. So, we chose these proteins to build the risk score for MCI and AD, respectively. Next, we arranged 36 MCI (**Figure S3**) and 50 AD (**Figure S4**) proteins by the average absolute value of their LASSO regression coefficients ($\bar{\beta}$). Association of individuals proteins with MCI and AD incidences and their aggregated PPRS performance were also presented in Figures S2 and S3.

1.2 Plasma AD biomarkers

1.2.1 Instrument and Software

- Simoa HD-1 Analyzer (Instrument ID: 27100100020; software version of 1.5)
- All samples was tested on the same instrument during the study.

1.2.2 Sample Handling

- The mixed samples were quickly spun for 30s using a bench-top mini centrifuge to remove any liquid from the caps. A proper volume of the samples (100 μ L of samples run in singlet; 130 μ L of samples run in duplicate) were then transferred to 1.7-mL micro-centrifuge tubes pre-labeled with barcodes that match the original sample tubes respectively.
- The transferred samples were then centrifuged at 20,000g for 3 minutes at 4C.
- After centrifugation, the samples were transferred to 96-well plates following a pre-defined plate map for testing. All samples were diluted 4-fold automatically on the HD-1 Analyzer using sample diluent (based on recommended dilution scheme in kit package insert).
- Original samples were immediately refrozen after the testing aliquot was removed.

1.2.3 Preparation of Calibrators and Controls

- Ready-to-use calibrators were manufactured and frozen (-80C) at the following Tau concentrations: 0, 0.093, 0.298, 1.04, 3.00, 9.38, 30.5, 95.1 pg/mL.
- Ready-to-use controls with assigned ranges were pre-manufactured and stored frozen at -80C.
- The same lot of calibrators and controls were used during this study.
- Controls were treated as samples and diluted 4-fold automatically on the HD-1 Analyzer.

1.2.4 Assay characteristics

Assay characteristics	p-tau	t-tau	A β ₄₀	A β ₄₂
Number of measurements (single/duplicate)	12% run in duplicate	11.6% in duplicate	9% in duplicate	9% in duplicate
Intra-assay CV	5.10%	4.10%	3.20%	2.60%

Inter-assay CV	9.60%	7.50%	10.50%	7.60%
Phantom samples	ICC = 0.992; CV=7.9%; N=130	ICC = 0.741; CV = 9.2%; N = 324	ICC = 0.916; CV = 4.8%; N = 146	ICC = 0.943; CV = 3.5%; N = 146

Table S1: List of 36 MCI proteins involved in MCI PPRS. Highlighted in blue are five proteins common in both MCI and AD PPRS.

S. No.	Protein description FHS	Symbol	S. No.	Protein description FHS	Symbol
1	Tissue-type plasminogen activator	TPA	19	Cell adhesion molecule 3	CADM3
2	Ectonucleoside triphosphate diphosphohydrolase 5	ENTPD5	20	Macrophage metalloelastase	MMP12
3	Roundabout homolog 2	ROBO2	21	Hepatocyte growth factor activator	HGFA
4	Proteasome subunit alpha type-2	PSMA2	22	Chromobox protein homolog 5	CBX5
5	Dickkopf-related protein 3	DKK3	23	Lactotransferrin	LTF
6	Extracellular matrix protein 1	ECM1	24	Desmocollin-3	DSC3
7	Matrix extracellular phosphoglycoprotein	MEPE	25	Pancreatic hormone	PPY
8	Chordin-like protein 1	CHRD1	26	Dynein light chain roadblock-type 1	DYNLRB1
9	Lymphotoxin-alpha	TNFB	27	Ligand-dependent nuclear receptor corepressor-like protein	LCORL
10	Interleukin-1 beta	IL1B	28	Chitotriosidase-1	CHIT1
11	Malate dehydrogenase, cytoplasmic	MDH1	29	40S ribosomal protein SA	RPSA
12	C-C motif chemokine 15	CCL15	30	Cyclin-dependent kinase inhibitor 1B	CDKN1B
13	Mast/stem cell growth factor receptor Kit	KIT	31	Complement C3	C3
14	Kunitz-type protease inhibitor 1	SPINT1	32	Advanced glycosylation end product-specific receptor, soluble	AGER
15	Cytochrome c	CYCS	33	Collectin-11	COLEC11
16	C-C motif chemokine 18	CCL18	34	Macrophage-capping protein	CAPG
17	Eotaxin	CCL11	35	gp41 C34 peptide, HIV	C34
18	Activin receptor type-1B	ACVR1B	36	NudC domain-containing protein 3	NUDCD3

Table S2: List of 50 AD proteins involved in AD PPRS. Highlighted in blue are five proteins common in both MCI and AD PPRS.

S. No.	Protein description FHS	Symbol	S. No.	Protein description FHS	Symbol
1	Alpha-2-antiplasmin	SERPINF2	26	BMP-binding endothelial regulator protein	BMPER
2	Mast/stem cell growth factor receptor Kit	KIT	27	Myeloblastin	PRTN3
3	A disintegrin and metalloproteinase with thrombospondin motifs 15	ADAMTS15	28	C-C motif chemokine 14	CCL14
4	Vascular endothelial growth factor receptor 2	KDR	29	Tumor necrosis factor-inducible gene 6 protein	TNFAIP6
5	Coiled-coil domain-containing protein 80	CCDC80	30	Polymeric immunoglobulin receptor	PIGR
6	Kallikrein-8	KLK8	31	L-Selectin	SELL
7	Pleiotrophin	PTN	32	Macrophage scavenger receptor types I and II	MSR1
8	Intercellular adhesion molecule 2	ICAM2	33	C-X-C motif chemokine 6	CXCL6
9	Fibrinogen	FGA	34	A disintegrin and metalloproteinase with thrombospondin motifs 5	ADAMTS5
10	cGMP-specific 3,5-cyclic phosphodiesterase	PDE5A	35	ATP synthase subunit beta, mitochondrial	ATP5F1B
11	Macrophage metalloelastase	MMP12	36	G2/mitotic-specific cyclin-B1	CCNB1
12	Death-associated protein kinase 2	DAPK2	37	Delta-like protein 4	DLL4
13	Spondin-1	SPON1	38	Human Chorionic Gonadotropin	CGB5
14	Ephrin type-B receptor 6	EPHB6	39	Chitotriosidase-1	CHIT1
15	Dual 3,5-cyclic-AMP and -GMP phosphodiesterase 11A	PDE11A	40	High affinity immunoglobulin gamma Fc receptor I	FCGR1A
16	Tyrosine-protein phosphatase non-receptor type 2	PTPN2	41	Phospholipase A2	PLA2G4A
17	Advanced glycosylation end product-specific receptor, soluble	AGER	42	Cyclin-dependent kinase 2:Cyclin-A2 complex	CDK2
18	Ephrin type-A receptor 1	EPHA1	43	Sialic acid-binding Ig-like lectin 6	SIGLEC6
19	Epidermal growth factor receptor	EGFR	44	Activated Protein C	APC
20	Tumor necrosis factor receptor superfamily member 27	EDA2R	45	Hepatocyte growth factor activator	HGFA
21	Toll-like receptor 2	TLR2	46	Immunoglobulin E	IgE
22	Complement C4	C4	47	C-X-C motif chemokine 11	CXCL11
23	Interleukin-22 receptor subunit alpha-1	IL22RA1	48	Cathepsin H	CTSH
24	Follicle stimulating hormone	FSHB	49	Integrin alpha-V: beta-5 complex	ITGAV
25	Annexin A1	ANXA1	50	Creatine kinase B-type heterodimer	CKMB

Table S3: Baseline characteristics of brain volume variables, cognitive domains and plasma AD biomarkers.

Characteristics ¹	Overall	CN	MCI	AD	P ²
MRI	n = 987	n = 809	n = 100	n = 78	
Time to MRI from protein	8.29 (2.76)	8.33 (2.81)	8.14 (2.69)	8.08 (2.29)	0.636
Hippocampal	0.54 (0.05)	0.54 (0.05)	0.53 (0.05)	0.51 (0.05)	5.23e-08
Total brain	7.26 (0.10)	7.27 (0.09)	7.27 (0.10)	7.24 (0.10)	0.159
Total gray	48.99 (2.11)	49.17 (2.07)	48.25 (2.02)	48.07 (2.20)	6.46e-08
Total white	39.93 (2.69)	40.15 (2.63)	39.45 (2.52)	38.22 (2.82)	1.12e-09
Temporal lobe	16.04 (0.85)	16.15 (0.82)	15.68 (0.79)	15.37 (0.89)	1.17e-18
Parietal lobe	15.30 (0.78)	15.35 (0.75)	15.08 (0.86)	14.99 (0.76)	5.99e-06
Total CSF	27.72 (2.89)	27.36 (2.79)	28.83 (2.73)	30.08 (2.67)	1.12e-18
Third ventricle	-7.13 (0.38)	-7.17 (0.37)	-7.03 (0.37)	-6.90 (0.33)	4.40e-10
Cognitive factors (close to exam 5)	n = 932	n = 736	n = 103	n = 93	
Memory	0.45 (0.47)	0.52 (0.41)	0.29 (0.38)	0.03 (0.68)	4.76e-25
Language	0.33 (0.46)	0.39 (0.41)	0.15 (0.47)	0.01 (0.58)	5.15e-18
Executive function	0.24 (0.54)	0.34 (0.48)	0.00 (0.48)	-0.30 (0.67)	2.08e-32
MMSE	n = 1114	n = 914	n = 113	n = 87	
Exam 05	29.04 (1.27)	29.16 (1.15)	28.56 (1.46)	28.46 (1.78)	3.86e-10
Exam 06	28.88 (1.39)	28.97 (1.30)	28.56 (1.52)	28.40 (1.94)	3.84e-05
Exam 07	28.90 (1.44)	29.03 (1.25)	28.36 (1.57)	28.14 (2.43)	2.54e-11
Exam 08	28.60 (2.03)	28.91 (1.46)	27.77 (1.96)	26.43 (4.35)	3.46e-33
p-tau181 (exam 9)	n = 843 2.58 (1.11)	n = 736 2.47 (1.06)	n = 65 3.08 (1.20)	n = 42 3.65 (1.14)	6.72e-14
t-tau (exam 8)	n = 836 1.78 (0.84)	n = 725 1.75 (0.84)	n = 70 2.01 (0.87)	n = 41 1.97 (0.77)	0.015
Aβ ₄₀ (exam 7)	n = 1306 5.05 (0.24)	n = 1075 5.04 (0.24)	n = 119 5.05 (0.20)	n = 112 5.09 (0.29)	0.120
Aβ ₄₂ (exam 7)	0.74 (0.04)	0.75 (0.04)	0.74 (0.04)	0.74 (0.04)	0.019

¹Mean (SD); n (%) ²One-way ANOVA; Abbreviations: CN, cognitive normal; MCI, mild cognitive impairment; AD, Alzheimer's disease; MMSE, Mini-Mental State Examination.

Table S4: Association of PPRS with MCI and AD incidences using Cox regression model adjusted for FHS CVD Risk score.

Outcome	Models	Covariate	β	HR (95% CI)	p	C-index (%)
MCI (n=1,367 and n_event=127)	MCI PPRS adjusted for CVD Risk score	MCI PPRS	1.71	5.52 (4.04, 7.53)	6e-27	85.63
		Age	0.08	1.09 (1.06, 1.12)	2E-08	
		Sex (female)	-0.09	0.92 (0.63, 1.34)	0.66	
		Education	-0.01	0.99 (0.92, 1.06)	0.8	
		ApoE ε4 (1)	0.38	1.46 (0.97, 2.22)	0.07	
		CVD Risk score	-0.05	0.95 (0.91, 1.00)	0.06	
AD (n=1,365 and n_event=125)	AD PPRS adjusted for CVD Risk score	AD PPRS	1.30	3.68 (2.79, 4.86)	4e-20	91.32
		Age	0.12	1.13 (1.09, 1.17)	2E-11	
		Sex (female)	-0.38	0.68 (0.43, 1.07)	0.1	
		Education	-0.02	0.98 (0.91, 1.06)	0.68	
		ApoE ε4 (1)	0.99	2.69 (1.84, 3.94)	4E-07	
		CVD Risk score	-0.01	0.99 (0.94, 1.04)	0.67	

Abbreviations: MCI, Mild cognitive impairment; AD, Alzheimer's disease; PPRS, plasma protein risks score; Beta, regression coefficient; HR, hazard ratio; CI, 95% confidence interval for HR; CVD, cardio vascular disease n, sample size; n_event, number of event.

Table S5: Association of MCI and AD specific proteins with age and MCI/AD incidence.

	Protein	Linear association with age			Cox PH for MCI or AD Incidence			
		β	Se(β)	<i>p</i>	β	HR	<i>p</i>	<i>p.adjust</i>
MCI (n=1387 and n_event= 129)	MEPE	-0.98	0.97	0.31	-1.00	0.37	0.0087	0.013
	HGFA	0.60	0.41	0.15	0.72	2.05	0.002	0.004
	CDKN1B	-0.53	0.42	0.21	0.25	1.29	0.03	0.035
	LCORL	0.02	0.93	0.98	-1.17	0.31	0.03	0.035
	CBX5	-0.17	1.01	0.86	-1.03	0.36	0.02	0.025
	DSC3	-0.87	0.60	0.15	-0.88	0.41	0.0099	0.013
AD (n=1386 and n_event=128)	SIGLEC6	0.73	0.71	0.30	0.39	1.47	0.04	0.041
	FGA	-0.03	1.74	0.99	-1.16	0.31	0.0095	0.01
	ICAM2	-1.91	1.20	0.11	-2.11	0.12	0.00014	0.0003
	DLL4	-0.84	0.72	0.24	-0.76	0.47	0.02	0.021
	HGFA	0.75	0.42	0.08	0.55	1.73	0.01	0.011
	MSR1	0.14	0.86	0.87	0.52	1.68	0.03	0.03
	APC	0.41	0.66	0.53	0.48	1.62	0.0002	0.0004
	IgE	-0.24	0.18	0.19	-0.22	0.80	0.00066	0.001
	DAPK2	0.34	0.83	0.69	0.74	2.09	0.00022	0.00043
	C4	-1.52	0.98	0.12	-0.79	0.46	0.00021	0.00043
	ITGAV	0.49	0.48	0.31	0.28	1.32	0.02	0.022
	PDE5A	-0.35	0.66	0.60	0.70	2.01	0.00051	0.00089
	CCNB1	0.12	0.61	0.84	0.31	1.37	0.03	0.03133

Abbreviations: MCI, mild cognitive impairment; AD, Alzheimer's disease; HR, hazard ratio; p.adjust, is the adjusted *p* value based on Benjamini & Hochberg method; n, sample size; n_event, number of events.

Figure S1: Participant selection flow chart and study design. p=the number of proteins, n = the number of subjects, CN = cognitive normal, MCI = mild cognitive impairment, AD = Alzheimer's disease.

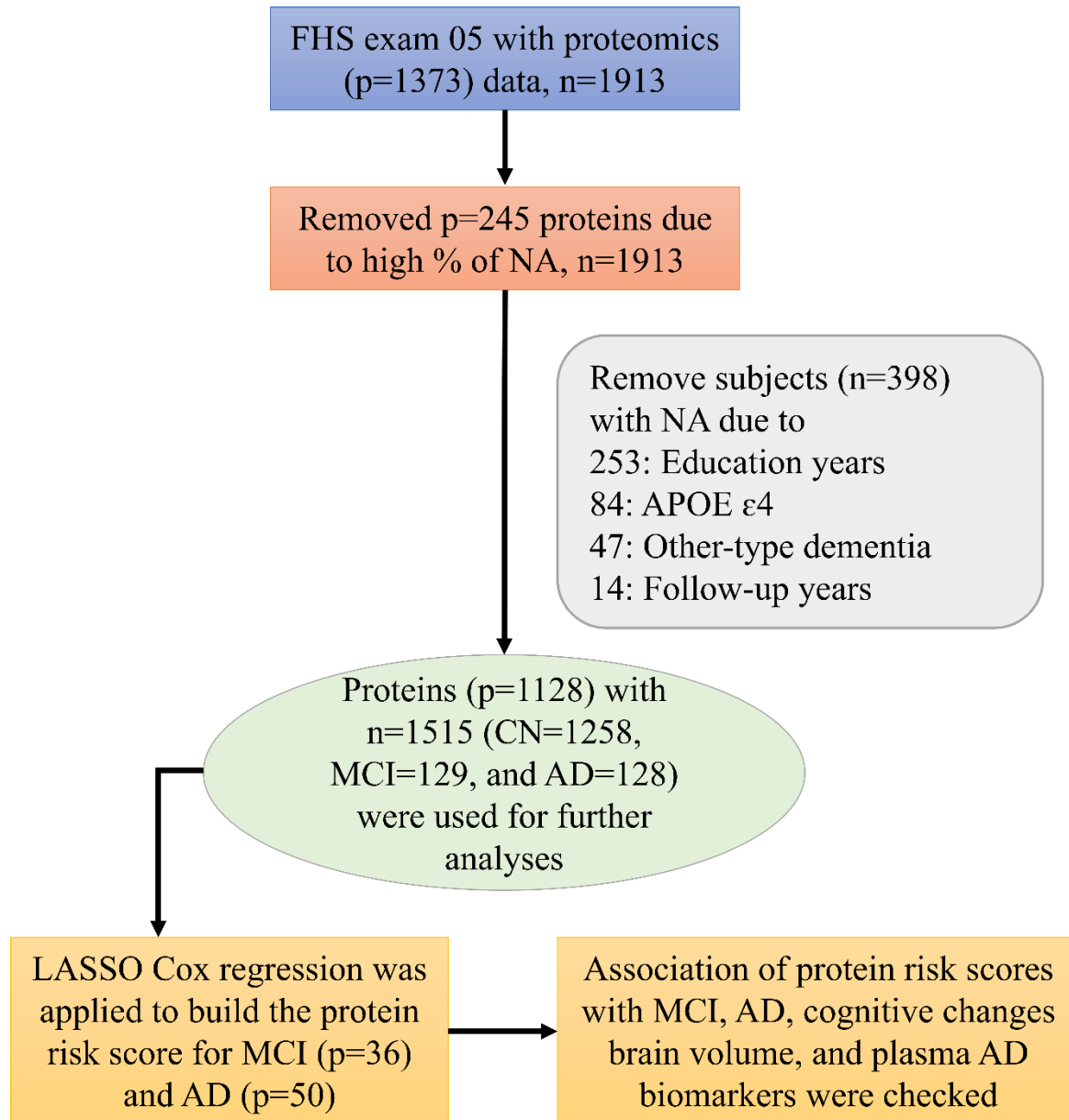


Figure S2: Plot for the difference in C-index for both MCI (**left panel**) and AD (**right panel**) proteins.

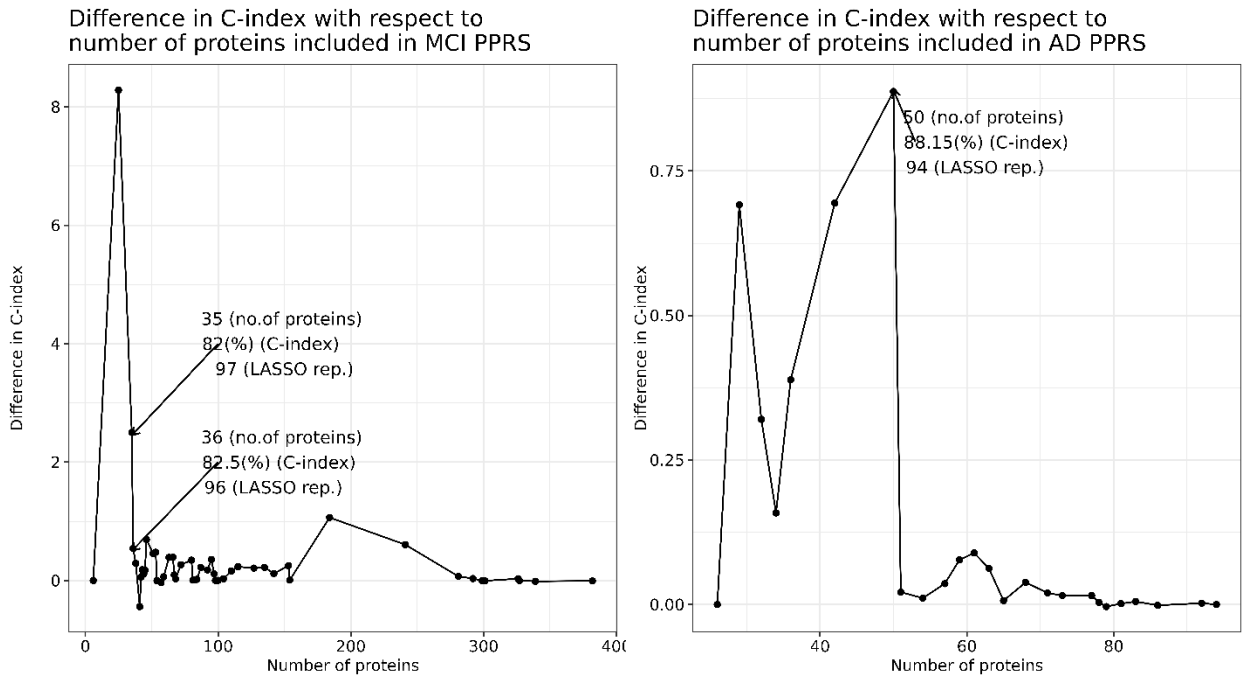


Figure S3: MCI proteins importance (36 proteins selected in Figure S1) plot and MCI PPRS performance. **(A)** Each protein association with MCI incidence using unadjusted Cox model and LASSO ($\bar{\beta}$) is average penalized regression coefficient which is used to calculate protein risk score. Proteins arranged based on absolute value of LASSO ($\bar{\beta}$). **(B)** MCI PPRS performance based on proteins included in MCI PPRS from top to bottom (1, 2, ... and so on) in panel (A). Final column denotes the protein is selected in final risk score or not based on p value (p) in panel (A). Age, sex, education, APOE ϵ 4, and CVD risk score were included as covariate. MCI, mild cognitive impairment; PPRS, plasma protein risk score.

	Protein unadjusted				B	MCI PPRS adjusted for covariates		
	β	HR (95% CI)	p	LASSO ($\bar{\beta}$)		HR (95% CI)	C-index	p
TPA	1.2662	3.55 (2.16, 5.83)	6.1e-07	0.82		3.26 (1.58, 6.74)	80.2	0.0014
ENTPD5	-1.3395	0.26 (0.12, 0.56)	0.00053	-0.72		4.82 (2.48, 9.38)	80.76	3.5e-06
ROBO2	-1.9157	0.15 (0.05, 0.4)	2e-04	-0.52		4.39 (2.41, 8)	80.84	1.4e-06
PSMA2	0.594	1.81 (0.92, 3.56)	0.08	0.49		5 (2.78, 9.01)	80.84	8.1e-08
DKK3	0.4654	1.59 (0.77, 3.31)	0.21	0.46		5.2 (2.83, 9.55)	80.94	1.1e-07
ECM1	-0.9544	0.39 (0.16, 0.95)	0.04	-0.43		4.8 (2.73, 8.46)	81.07	5.7e-08
MEPE	-0.9977	0.37 (0.17, 0.78)	0.0087	-0.42		5.71 (3.28, 9.94)	81.22	7.7e-10
CHRD1	1.008	2.74 (1.33, 5.65)	0.0063	0.4		5.75 (3.25, 10.16)	81.34	1.8e-09
TNFB	-0.5526	0.58 (0.27, 1.22)	0.15	-0.37		5.89 (3.37, 10.29)	81.73	4.7e-10
IL1B	0.5802	1.79 (1.32, 2.41)	0.00015	0.35		5.27 (3.35, 8.3)	82.14	6.8e-13
MDH1	-1.1618	0.31 (0.14, 0.68)	0.0032	-0.34		4.97 (3.21, 7.71)	82.22	7.6e-13
CCL15	-0.4771	0.62 (0.37, 1.05)	0.08	-0.33		4.76 (3.16, 7.17)	82.29	9.3e-14
KIT	-1.1473	0.32 (0.19, 0.54)	1.9e-05	-0.33		4.56 (3.08, 6.75)	82.32	3.6e-14
SPINT1	-1.0083	0.36 (0.22, 0.61)	0.00013	-0.31		4.12 (2.86, 5.92)	82.32	2.2e-14
CYCS	0.4139	1.51 (1.08, 2.11)	0.01	0.31		4.38 (3.07, 6.25)	82.46	4.7e-16
CCL18	0.6813	1.98 (1.3, 3)	0.0013	0.3		4.79 (3.35, 6.84)	82.79	8.2e-18
CCL11	0.4696	1.6 (0.98, 2.62)	0.06	0.3		5.74 (3.91, 8.42)	83.19	4.2e-19
ACVR1B	0.6205	1.86 (1.3, 2.66)	0.00072	0.27		5.31 (3.69, 7.64)	83.16	2.4e-19
CADM3	-0.8085	0.45 (0.24, 0.82)	0.0092	-0.25		4.98 (3.53, 7.03)	83.38	6.7e-20
MMP12	0.7536	2.12 (1.44, 3.13)	0.00015	0.24		5.19 (3.65, 7.38)	83.44	4.4e-20
HGFA	0.7157	2.05 (1.3, 3.22)	0.002	0.21		5.04 (3.57, 7.13)	83.71	5.3e-20
CBX5	-1.0294	0.36 (0.16, 0.82)	0.02	-0.2		4.95 (3.52, 6.98)	83.78	5.9e-20
LTF	-0.6667	0.51 (0.34, 0.77)	0.0012	-0.19		4.42 (3.2, 6.11)	83.64	2.1e-19
DSC3	-0.8829	0.41 (0.21, 0.81)	0.0099	-0.19		4.4 (3.19, 6.08)	83.7	2.2e-19
PPY	0.2993	1.35 (1.01, 1.8)	0.04	0.18		5.07 (3.62, 7.11)	84.29	3.3e-21
DYNLRB1	-0.3637	0.7 (0.39, 1.23)	0.21	-0.17		5.39 (3.84, 7.57)	84.6	2.1e-22
LCORL	-1.1652	0.31 (0.11, 0.91)	0.03	-0.16		5.43 (3.87, 7.62)	84.64	1.4e-22
CHIT1	0.525	1.69 (1.25, 2.28)	0.00057	0.15		5.3 (3.79, 7.4)	84.47	1.4e-22
RPSA	-0.5019	0.61 (0.35, 1.04)	0.07	-0.14		5.39 (3.87, 7.52)	84.65	3.2e-23
CDKN1B	0.2538	1.29 (1.03, 1.61)	0.03	0.13		5.43 (3.94, 7.49)	84.92	5.6e-25
C3	-0.4445	0.64 (0.42, 0.99)	0.04	-0.12		5.35 (3.9, 7.33)	85.06	1.7e-25
AGER	-0.5248	0.59 (0.42, 0.84)	0.0037	-0.1		5.35 (3.91, 7.31)	85.19	7.4e-26
COLEC11	0.2359	1.27 (0.86, 1.86)	0.23	0.1		5.55 (4.05, 7.6)	85.45	1.3e-26
CAPG	0.4925	1.64 (1.13, 2.36)	0.0084	0.1		5.56 (4.07, 7.61)	85.53	7.2e-27
C34	0.5563	1.74 (1.23, 2.46)	0.0016	0.08		5.49 (4.02, 7.5)	85.55	9.3e-27
NUDCD3	-0.817	0.44 (0.21, 0.95)	0.04	-0.07		5.52 (4.04, 7.53)	85.63	5.8e-27

Figure S4: AD proteins importance (50 proteins selected in Figure S1) plot and AD PPRS performance. **(A)** Each protein association with AD incidence using unadjusted Cox model and LASSO ($\bar{\beta}$) is average penalized regression coefficient which is used to calculate protein risk score. Proteins arranged based on absolute value of LASSO ($\bar{\beta}$). **(B)** AD PPRS performance based on proteins included in AD PPRS from top to bottom (1, 2, ... and so on) in panel (A). Final column denotes the protein is selected in final risk score or not based on p value (p) in panel (A). Age, sex, education, APOE ϵ 4, and CVD risk score were included as covariate. AD, Alzheimer's disease; PPRS, plasma protein risk score.

	Protein unadjusted				AD PPRS adjusted for covariates		
	β	HR (95% CI)	p	LASSO ($\bar{\beta}$)	HR (95% CI)	C-index	p
SERPINF2	-1.1815	0.31 (0.2, 0.47)	2.6e-08	-1.05	1.82 (0.71, 4.66)	89.04	0.21
KIT	-1.6456	0.19 (0.11, 0.32)	6e-10	-0.72	2.39 (1.36, 4.19)	89.18	0.0024
ADAMTS15	0.9987	2.71 (1.66, 4.45)	7.4e-05	0.58	2.96 (1.9, 4.61)	89.5	1.6e-06
KDR	-2.2193	0.11 (0.05, 0.24)	2.8e-08	-0.54	2.8 (1.83, 4.29)	89.51	2.2e-06
CCDC80	1.6001	4.95 (2.2, 11.17)	0.00012	0.52	2.66 (1.7, 4.16)	89.43	2e-05
KLK8	-1.3999	0.25 (0.12, 0.49)	8e-05	-0.48	2.69 (1.78, 4.07)	89.46	2.4e-06
PTN	0.8614	2.37 (1.59, 3.51)	1.9e-05	0.48	2.07 (1.38, 3.11)	89.23	0.00043
ICAM2	-2.1126	0.12 (0.04, 0.36)	0.00014	-0.47	2.24 (1.52, 3.31)	89.42	5.1e-05
FGA	-1.1583	0.31 (0.13, 0.75)	0.0095	-0.4	2.36 (1.59, 3.5)	89.51	1.9e-05
PDE5A	0.6959	2.01 (1.35, 2.97)	0.00051	0.37	2.7 (1.86, 3.94)	89.77	2.2e-07
MMP12	1.0159	2.76 (1.89, 4.03)	1.3e-07	0.35	2.61 (1.78, 3.81)	89.66	7.5e-07
DAPK2	0.7362	2.09 (1.41, 3.09)	0.00022	0.34	2.93 (2.03, 4.23)	89.84	8.3e-09
SPON1	1.2618	3.53 (1.43, 8.7)	0.0061	0.33	2.86 (1.98, 4.14)	89.81	2.3e-08
EPHB6	-1.3724	0.25 (0.09, 0.69)	0.0071	-0.33	2.95 (2.04, 4.25)	89.85	8.1e-09
PDE11A	-1.7178	0.18 (0.07, 0.44)	0.00017	-0.29	2.92 (2.03, 4.19)	89.91	7.2e-09
PTPN2	0.7606	2.14 (1.39, 3.29)	0.00056	0.27	3.05 (2.14, 4.34)	90.02	5.7e-10
AGER	-0.6705	0.51 (0.36, 0.73)	0.00023	-0.25	3.27 (2.32, 4.62)	90.28	1.4e-11
EPHA1	0.4887	1.63 (0.99, 2.69)	0.06	0.24	3.48 (2.45, 4.93)	90.34	3e-12
EGFR	-2.2023	0.11 (0.05, 0.24)	1.6e-08	-0.23	3.34 (2.37, 4.71)	90.3	4.9e-12
EDA2R	0.6526	1.92 (1.38, 2.68)	0.00012	0.23	3.08 (2.24, 4.23)	90.31	3.5e-12
TLR2	-0.7387	0.48 (0.21, 1.07)	0.07	-0.2	3.36 (2.42, 4.68)	90.38	6.7e-13
C4	-0.7863	0.46 (0.3, 0.69)	0.00021	-0.19	3.36 (2.44, 4.65)	90.52	1.8e-13
IL22RA1	0.5969	1.82 (1.26, 2.61)	0.0012	0.19	3.5 (2.54, 4.84)	90.57	2.6e-14
F5HB	0.5032	1.65 (1.42, 1.92)	4.6e-11	0.17	3.55 (2.57, 4.91)	90.58	1.7e-14
ANXA1	1.043	2.84 (1.86, 4.34)	1.4e-06	0.17	3.42 (2.51, 4.67)	90.57	7.9e-15
BMPER	0.4136	1.51 (1.18, 1.94)	0.0011	0.16	3.42 (2.53, 4.63)	90.64	1.7e-15
PRTN3	-0.7955	0.45 (0.27, 0.75)	0.0021	-0.16	3.45 (2.55, 4.67)	90.67	8.9e-16
CCL14	1.1491	3.16 (1.7, 5.85)	0.00026	0.15	3.53 (2.6, 4.79)	90.73	6.7e-16
TNFAIP6	-1.2284	0.29 (0.17, 0.5)	5.8e-06	-0.15	3.64 (2.67, 4.97)	90.78	4.2e-16
PIGR	-0.6296	0.53 (0.34, 0.83)	0.0055	-0.14	3.62 (2.65, 4.96)	90.76	7.7e-16
SELL	-1.7547	0.17 (0.07, 0.4)	4.9e-05	-0.14	3.58 (2.63, 4.89)	90.77	8e-16
MSR1	0.5168	1.68 (1.06, 2.65)	0.03	0.12	3.58 (2.64, 4.86)	90.78	2.5e-16
CXCL6	0.3755	1.46 (1.14, 1.85)	0.0023	0.12	3.7 (2.72, 5.03)	90.81	6.1e-17
ADAMTS5	0.6218	1.86 (1.38, 2.51)	4.2e-05	0.11	3.65 (2.69, 4.96)	90.83	8.3e-17
ATPSF1B	0.5822	1.79 (1.16, 2.77)	0.009	0.11	3.62 (2.68, 4.89)	90.85	6.6e-17
CCNB1	0.3129	1.37 (1.03, 1.81)	0.03	0.11	3.57 (2.65, 4.82)	90.91	5.2e-17
DLL4	-0.7638	0.47 (0.24, 0.9)	0.02	-0.1	3.62 (2.69, 4.88)	90.98	3.1e-17
CGB5	0.4489	1.57 (1.37, 1.79)	6.4e-11	0.09	3.56 (2.64, 4.79)	90.98	5.2e-17
CHIT1	0.6265	1.87 (1.39, 2.52)	3.5e-05	0.09	3.55 (2.65, 4.77)	90.94	3.8e-17
FCGR1A	-0.4369	0.65 (0.37, 1.11)	0.12	-0.09	3.59 (2.67, 4.82)	90.95	2.3e-17
PLA2G4A	-0.6032	0.55 (0.3, 0.99)	0.05	-0.09	3.67 (2.72, 4.93)	91.01	9.7e-18
CDK2	0.7103	2.03 (1.35, 3.07)	0.00073	0.08	3.7 (2.75, 4.97)	91.09	4.2e-18
SIGLEC6	0.3879	1.47 (1.03, 2.12)	0.04	0.08	3.73 (2.78, 4.99)	91.14	9.7e-19
APC	0.4811	1.62 (1.26, 2.08)	2e-04	0.08	3.57 (2.69, 4.72)	91.12	7.9e-19
HGFA	0.548	1.73 (1.12, 2.66)	0.01	0.07	3.6 (2.72, 4.76)	91.19	4.1e-19
IgE	-0.2223	0.8 (0.7, 0.91)	0.00066	-0.07	3.7 (2.79, 4.89)	91.34	5e-20
CXCL11	0.4752	1.61 (1.21, 2.14)	0.0012	0.07	3.68 (2.78, 4.86)	91.33	6.1e-20
CTSH	2.8459	17.22 (4.87, 60.87)	1e-05	0.06	3.67 (2.78, 4.85)	91.33	6.2e-20
ITGAV	0.2769	1.32 (1.05, 1.66)	0.02	0.06	3.72 (2.82, 4.92)	91.32	2.8e-20
CKMB	-0.7418	0.48 (0.36, 0.63)	2.1e-07	-0.05	3.68 (2.79, 4.86)	91.32	3.9e-20

Figure S5: Forest plots for the results of the probability of an individual experiencing the MCI incident based on three different values of age (60, 70, and 80 years old(yo)) and MCI PPRS level (**left panel**) and the AD incident based on different values of age (60, 70, and 80 years old(yo)) and AD PPRS level (**right panel**). These probabilities were calculated at different follow-up time points, 10 years, 15 years, and 20 years, using Cox PH models. MCI, mild cognitive impairment; AD, Alzheimer's disease; PPRS, plasma protein risk score.

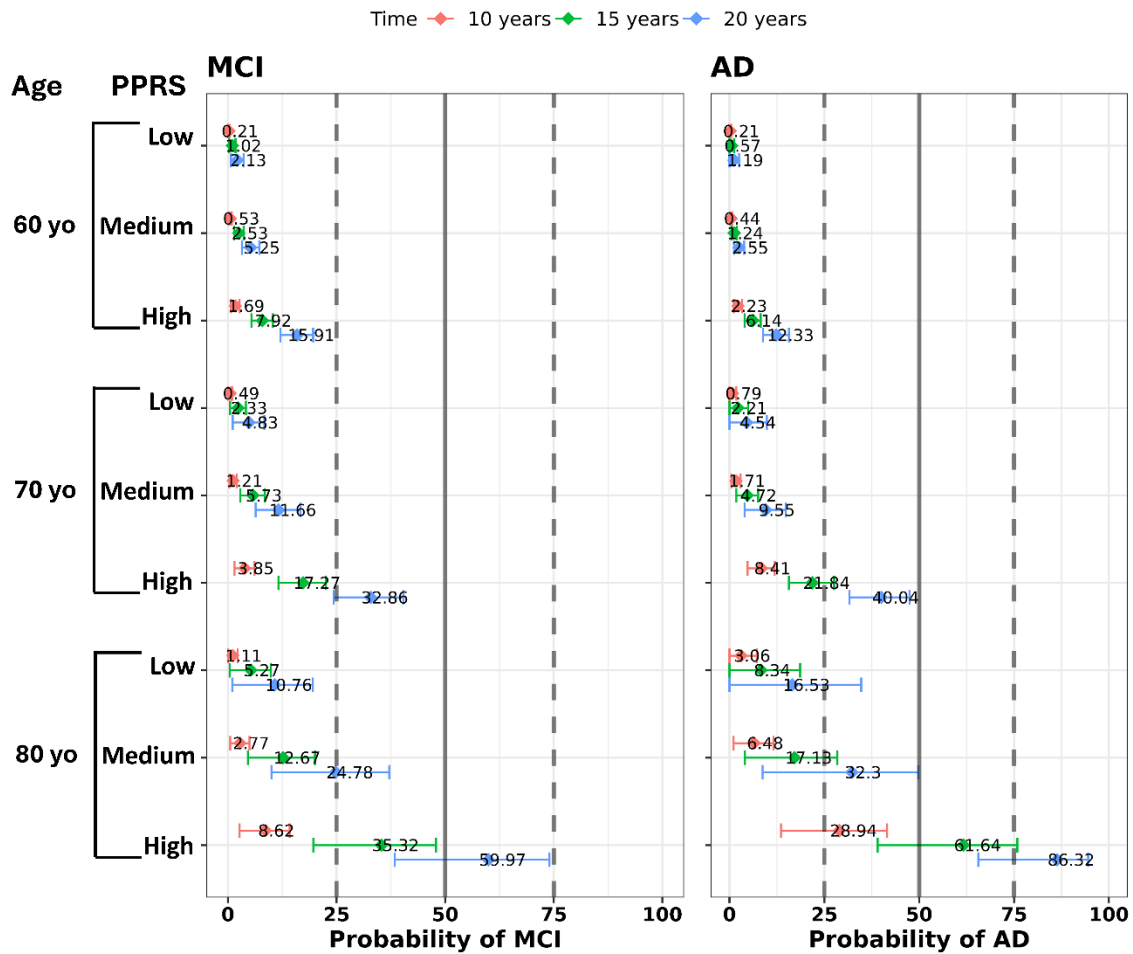


Figure S6: Association of MCI (A) and AD (B) PPRSs with MMSE at Exam 06-08. FHS Exams 05-08 were conducted consecutively approximately 4-6 years later. MMSE, Mini-Mental State Examination.

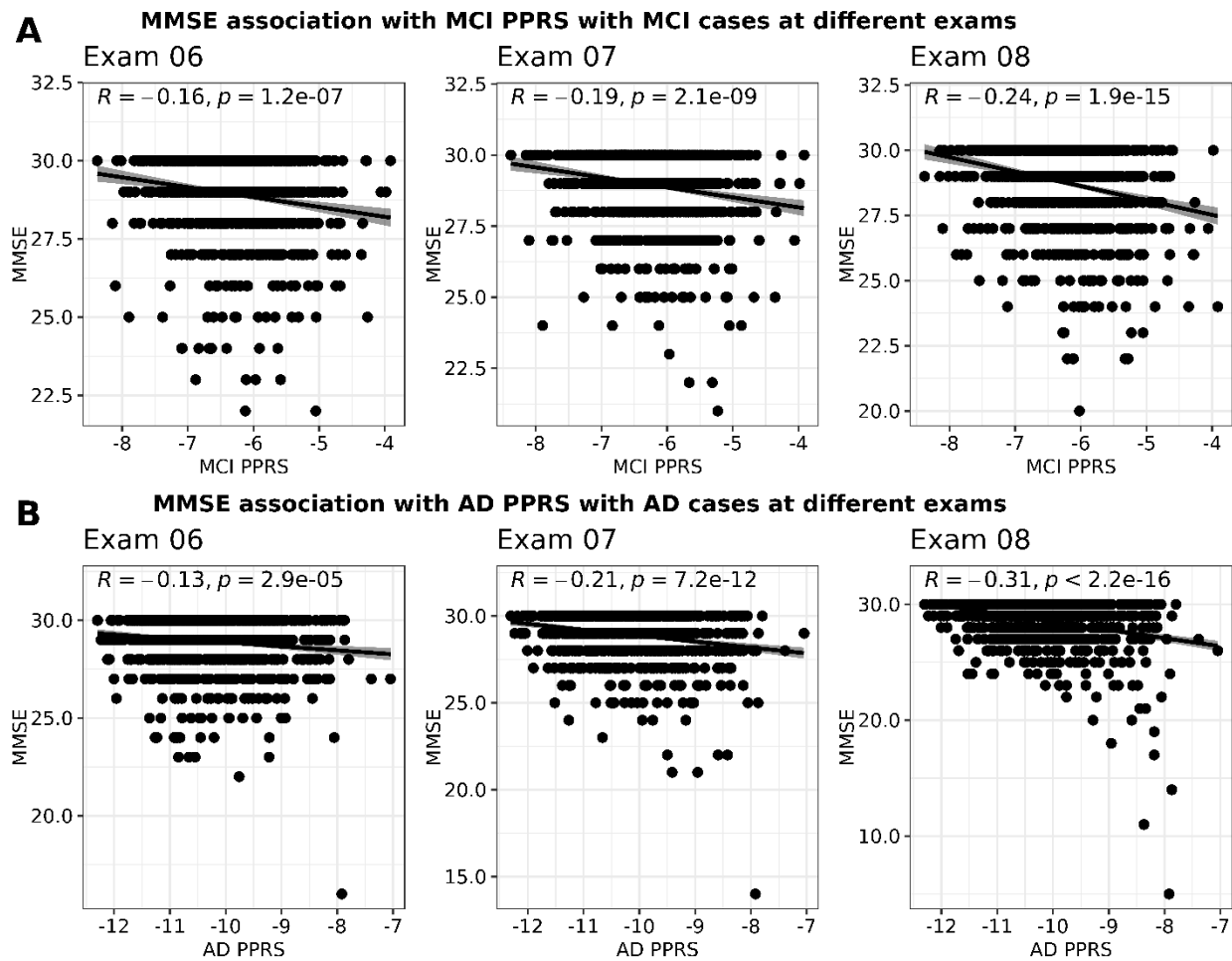


Figure S7: The association between MMSE and age in different groups of MCI (A) and AD (B) PPRSs on Exam 05-08. FHS Exams 05-08 were administered sequentially around 4-6 years later. MMSE, Mini-Mental State Examination.

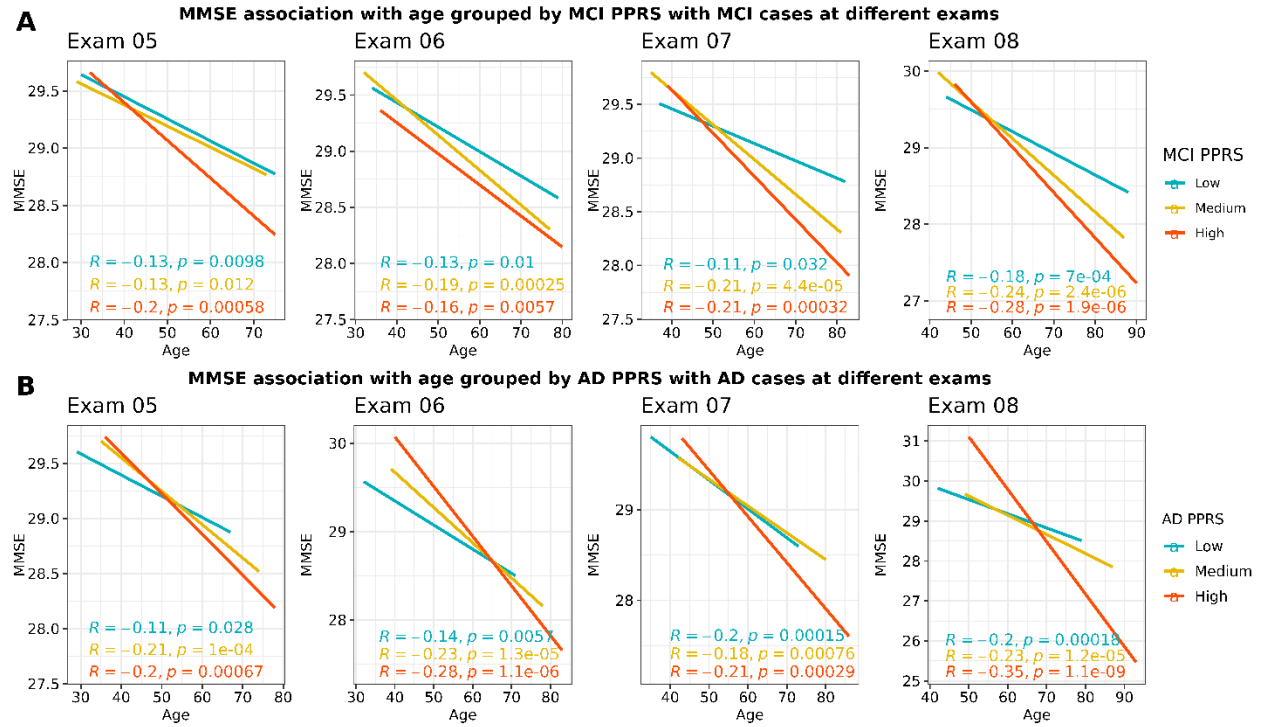


Figure S8: The association between three cognitive domains (memory, language, and executive function) and age in different groups of MCI and AD PPRSs. Memory, language, and executive function measurements were closest to Exam 05 (baseline).

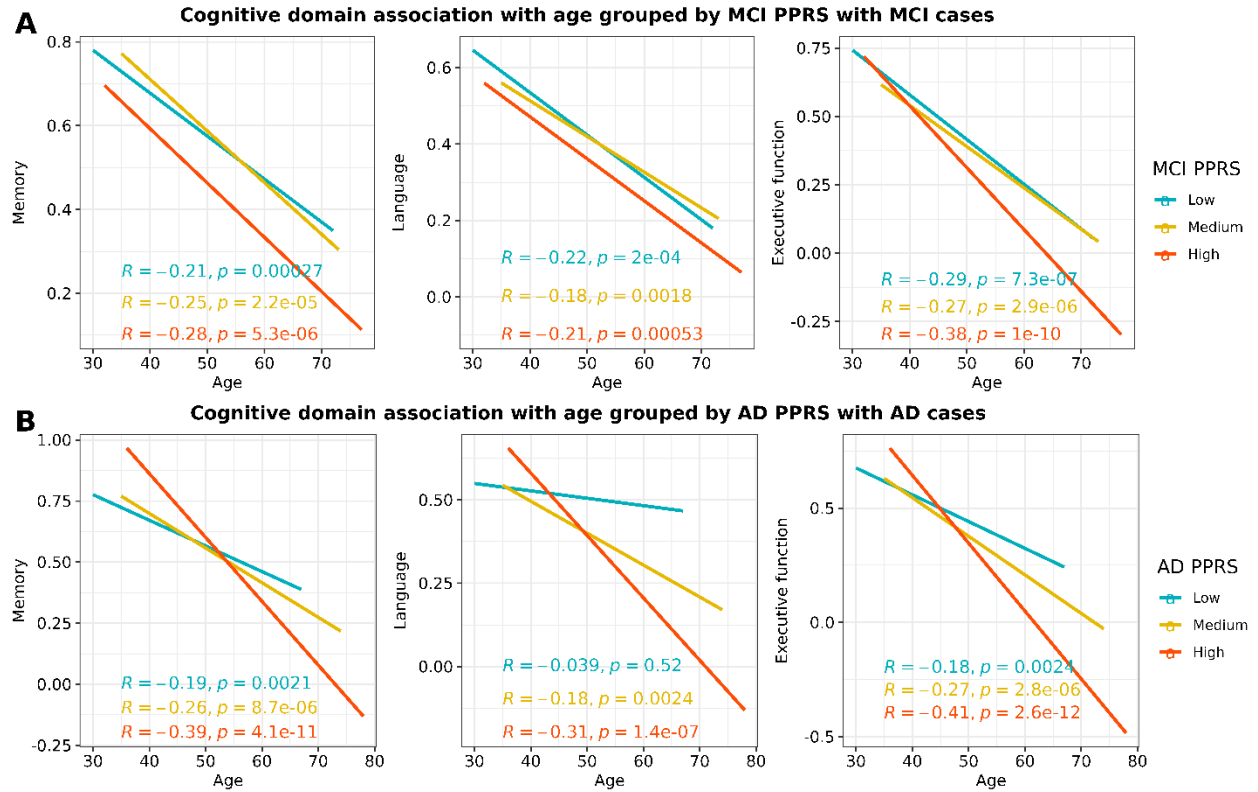


Figure S9: Brain volume distribution over (A) MCI PPRS and (B) AD PPRS for younger than 60 years participants. MCI, mild cognitive impairment; AD, Alzheimer's disease; PPRS, plasma protein risk score; CSF, cerebrospinal fluid.

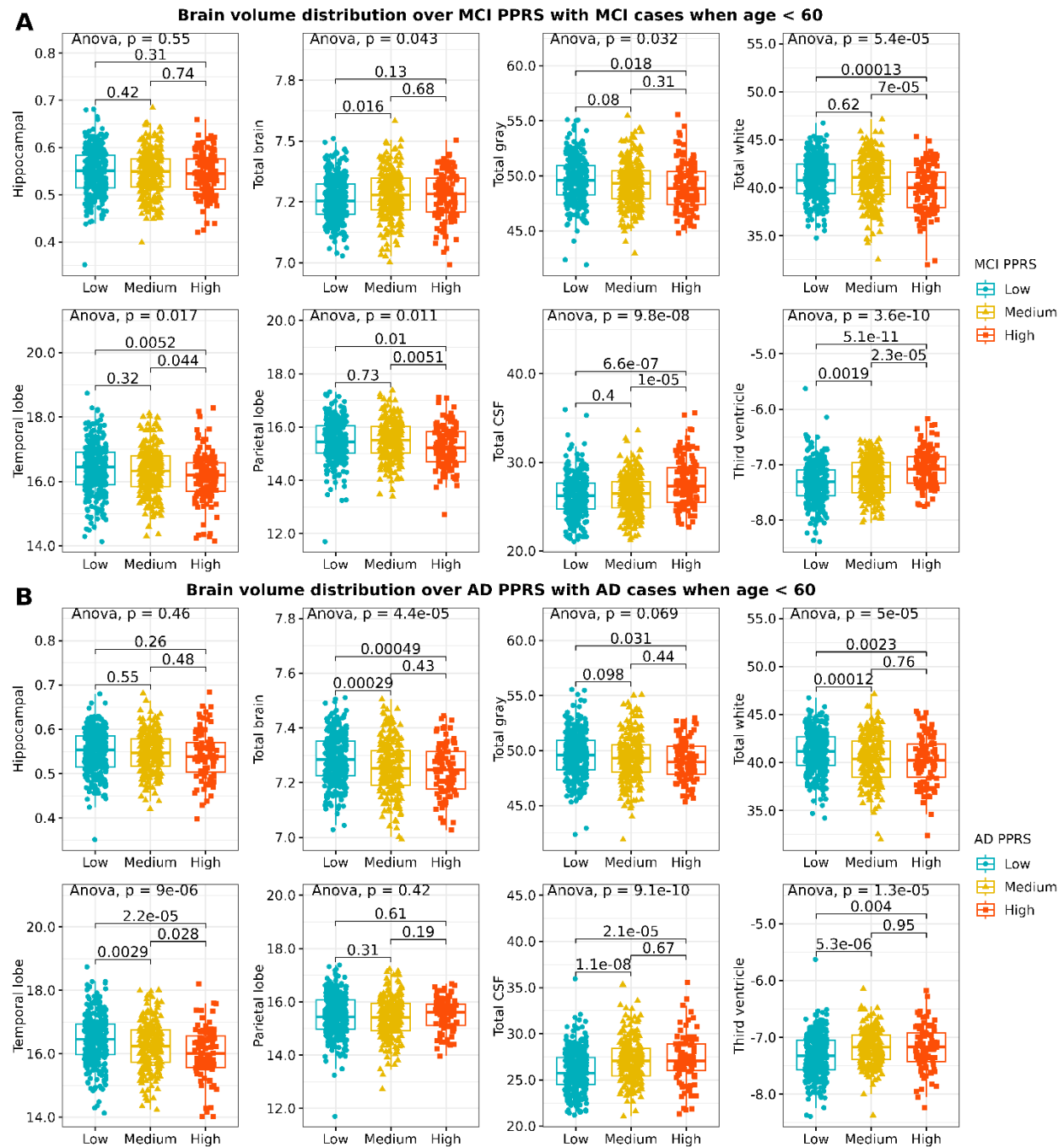


Figure S10: Brain volume distribution over (A) MCI PPRS and (B) AD PPRS without MCI and AD cases respectively, i.e. only CN participants. MCI, mild cognitive impairment; AD, Alzheimer's disease; PPRS, plasma protein risk score; CSF, cerebrospinal fluid.

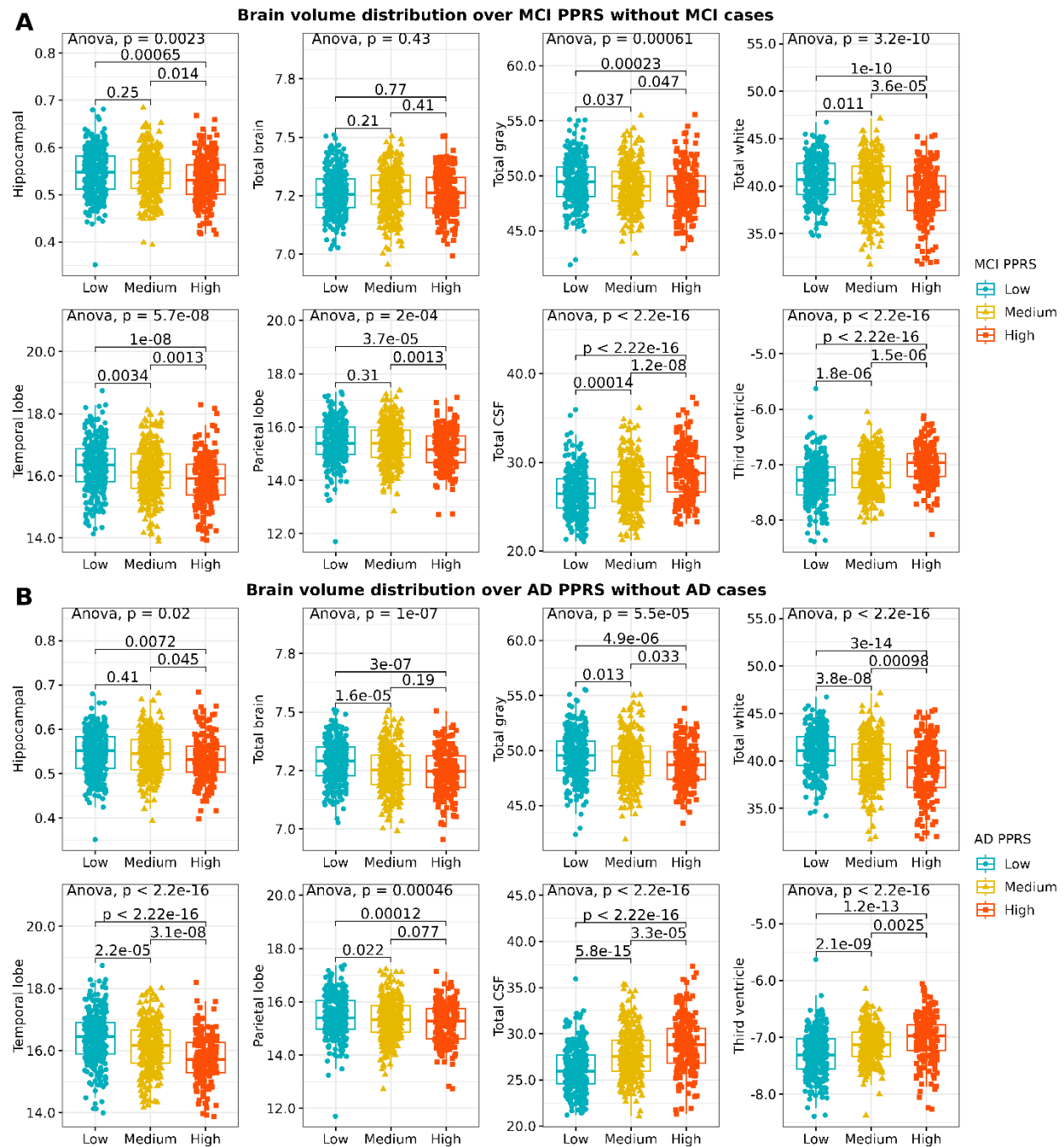


Figure S11: Plasma biomarkers distribution over (A) MCI PPRS and (B) AD PPRS for younger than 60 years participants. MCI, mild cognitive impairment; AD, Alzheimer's disease; PPRS, plasma protein risk score; CSF, cerebrospinal fluid.

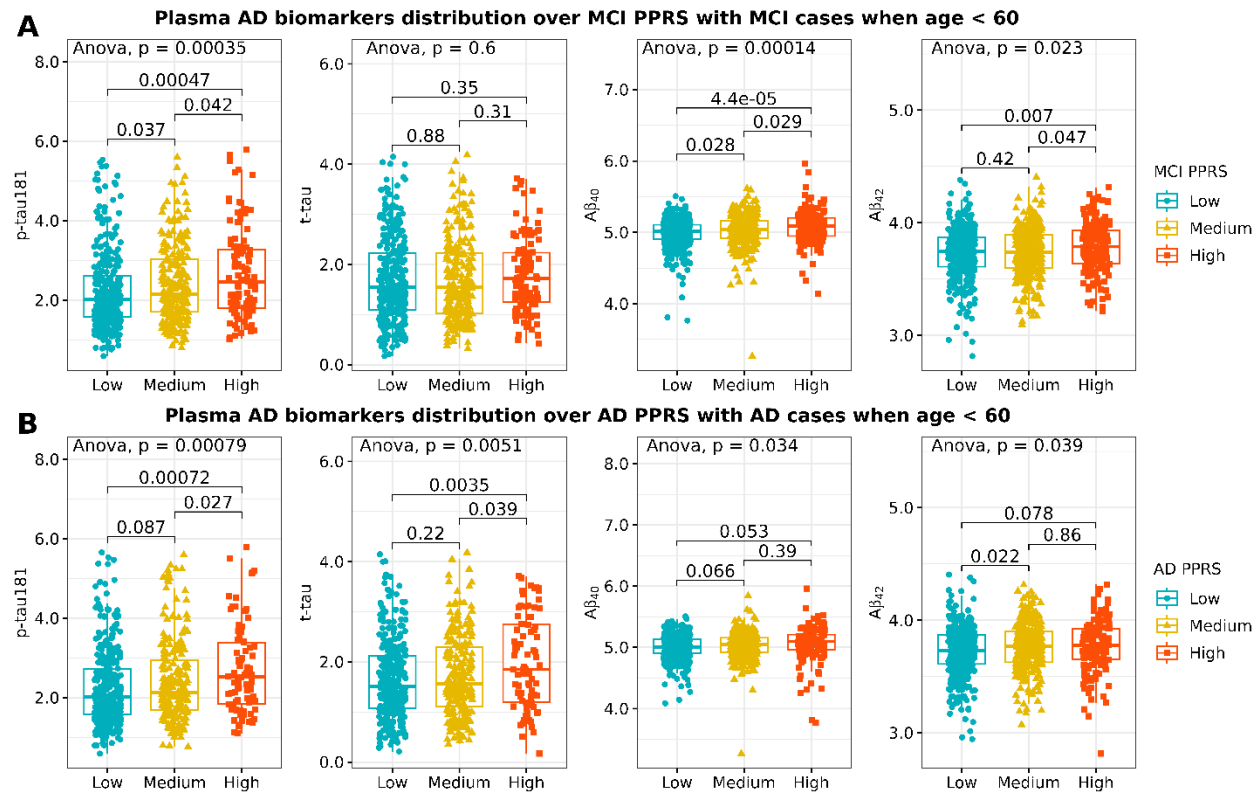


Figure S12: Plasma biomarkers distribution over (A) MCI PPRS and (B) AD PPRS without MCI and AD cases respectively, i.e. only CN participants. MCI, mild cognitive impairment; AD, Alzheimer's disease; PPRS, plasma protein risk score; CSF, cerebrospinal fluid.

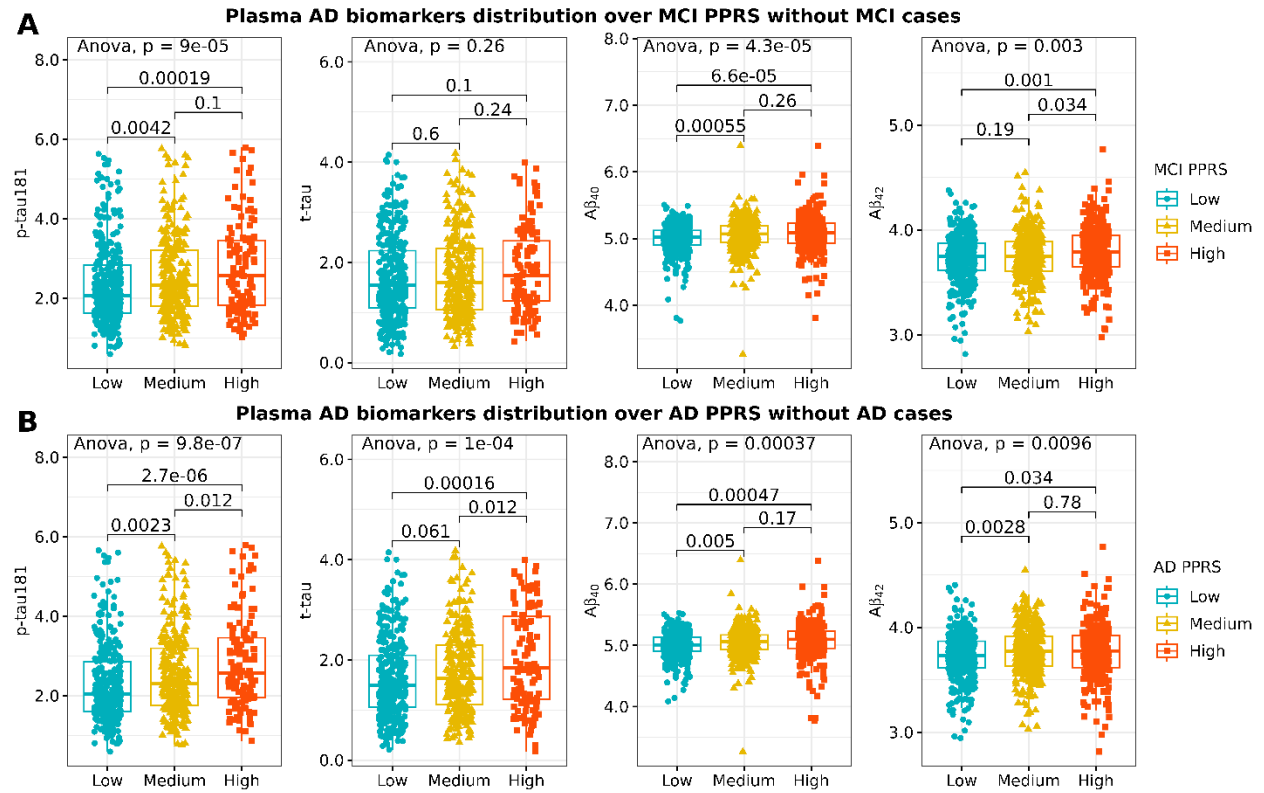


Figure S13: Comparison of expression level of 5 common proteins among CN, MCI, and AD individuals. CN, Cognitive normal; MCI, Mild cognitive impairment; AD, Alzheimer’s disease.

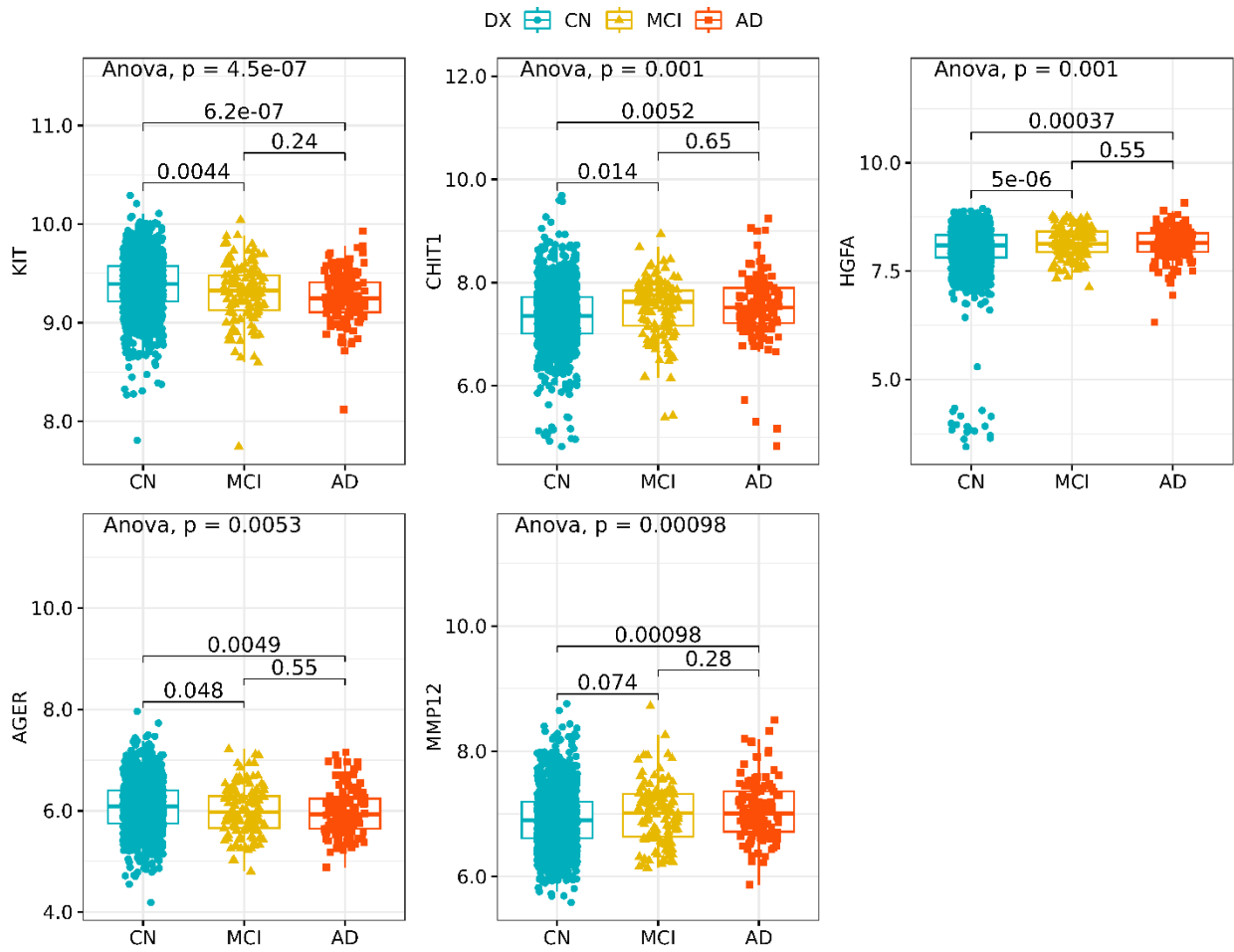


Figure S14: Validation in the ADNI CSF proteomics data set. (A) Association of 8 proteins and aggregated protein risk score (PPRS) with incidence of MCI from CN. PPRS was calculated using the sum of the product of regression coefficients (β) with respected protein expression levels. The FHS PPRS row indicates that proteins belong to MCI or AD PPRS in the FHS training set. (B) Comparison of low, medium, and high PPRS individuals based on CSF AD biomarkers and AD risk factors at baseline. PPRS in low, medium, and high were grouped based on tertiles. AV45 ratio of cortical grey matter and whole cerebellum. Summary florbetapir cortical SUVR normalized by whole cerebellum. CSF A β 42, p-tau181, t-tau and hippocampus volume were log transformed. MCI, mild cognitive impairment; AD, Alzheimer’s disease; PPRS, plasma protein risk score; FHS, Framingham Heart Study; HR, Hazard ratio; CDRSB, Clinical Dementia Rating scale Sum of Boxes; MMSE, Mini-Mental State Examination.

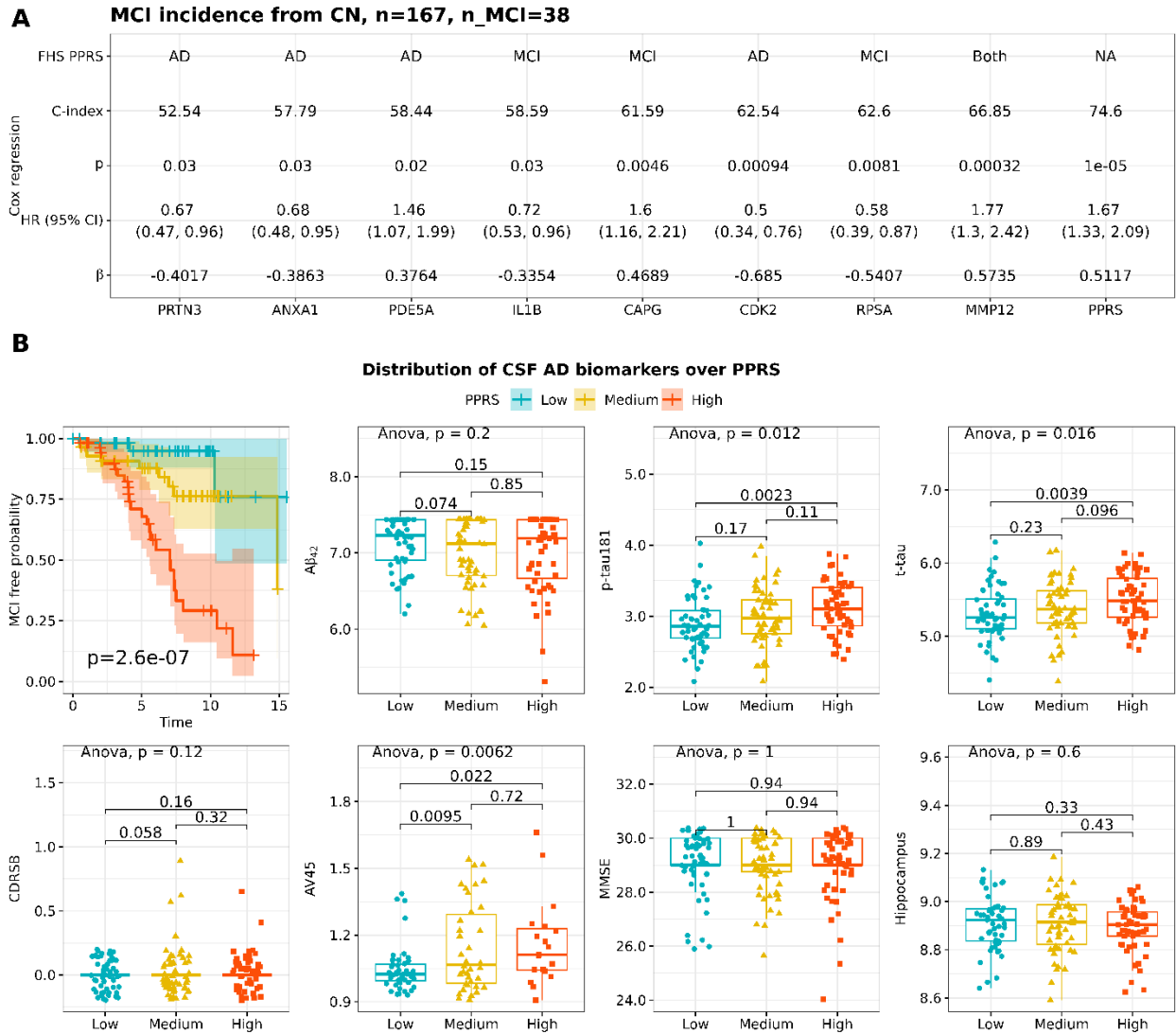


Figure S15: Performance of MCI and AD PPRSs in FHS test data set. **(A)** Cox regression results for MCI and AD PPRS unadjusted as well as AD PPRS adjusted for CVD risk score. **(B)** MCI free probability based on different levels (low and high) of MCI PPRS **(C)** AD free probability based on different levels (low and high) of AD PPRS in FHS testing data. PPRS were grouped based on median cutoff (less than median =low and greater than median=high). MCI, mild cognitive impairment; AD, Alzheimer’s disease; PPRS, plasma protein risk score; Adjus., adjusted.

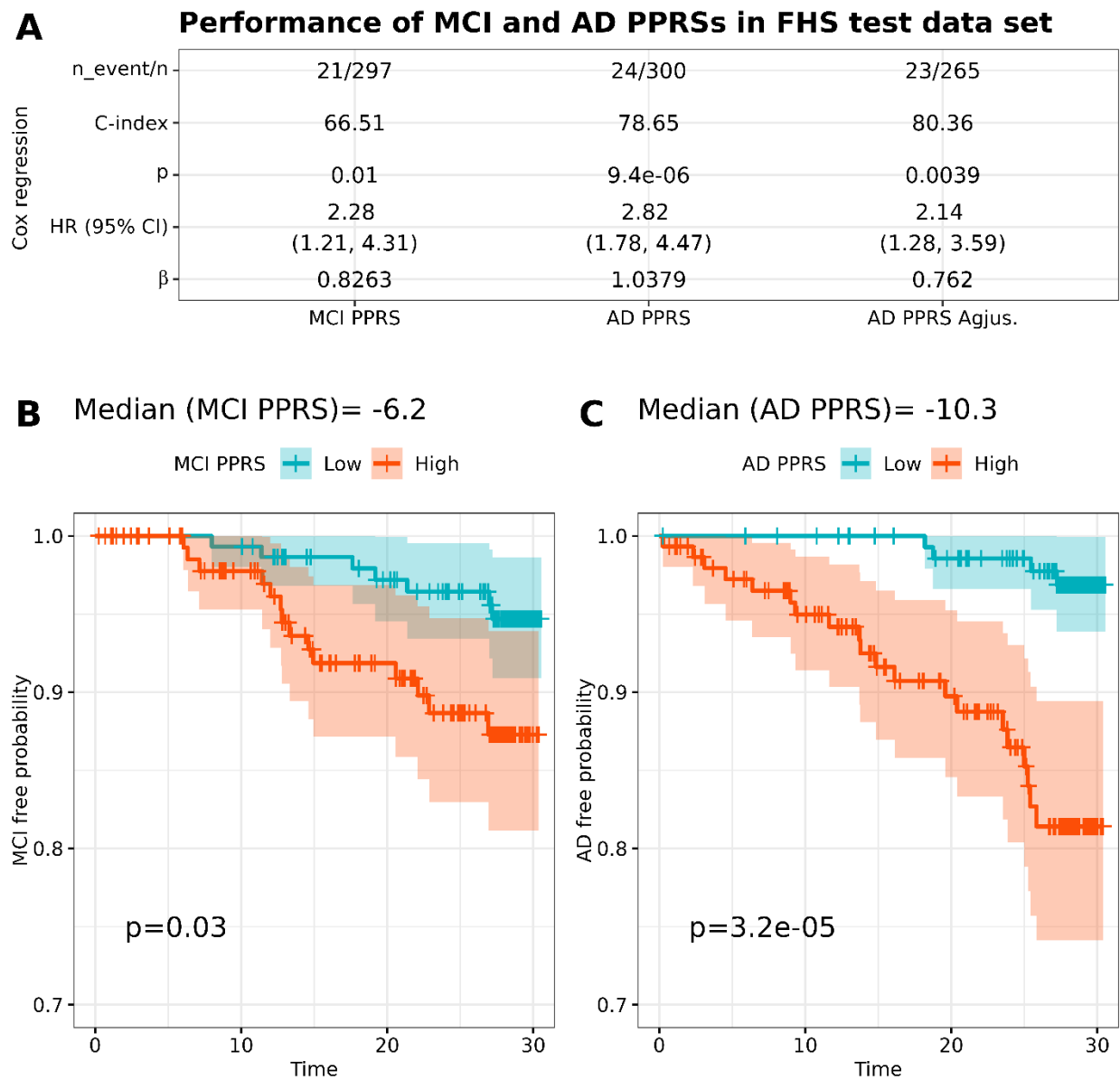


Figure S16: Distribution of (A) MCI PPRS and (B) AD PPRS among male and female.

